

Khawaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I

Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring - 2024	Date: 10.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 4
----------------------------	-------------------------	-----------------------	------------------	----------------------------

Topics to be discussed: Dipole in an electric field, electric flux, Gauss' law

Target CLO & PLO : CLO2-3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define electric flux and charge density
- ❖ State and explain Gauss's Law
- ❖ Explain how to derive Coulomb's Law from Gauss's Law
- ❖ Applications of Gauss's Law

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) What is the unit of electric flux?
a) Nm b) Vm c) Fm d) NC^{-1} e) CN^{-1}
- 2) Gauss law is related to _____ law?
a) Coulombs law b) Ohms law c) Charles law d) Farads law e) Amperes Law
- 3) If the radius of the spherical gaussian surface is double then the flux will –
a) be double. b) increase 4th times. c) be reduced to half.
d) be reduced to a quarter e) remain same.

Sample Questions (SAQ):

- 1) Define electric flux & charge density.
- 2) Explain the Gaussian surface.
- 3) State and explain Gauss's Law.
- 4) Deduce Coulomb's law from Gauss's law.
- 5) Using Gauss's law, calculate the electric field at a distance y from an infinite line of charge of with uniform charge density λ .
- 6) Prove that the net flux of a cylindrical Gaussian surface placed in an external electric field is zero.
- 7) An electric field of 500V/m makes an angle 60° with the surface vector which has a magnitude of 0.500m^2 . Find the flux that passes through the surface.
- 8) A point charge of $2.0\mu\text{C}$ is at the center of a cubic Gaussian surface 8.0cm edge. What the net electric flux through the surface?

Reference Books: 1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 332-342
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86